

Indian Institute of Technology Goa



RETINAL BLOOD FLOW SIMULATION

B.Tech Final Year Report

Submitted by:

Abhinav Yellamati

Roll No: 2203140

Department: Computer Science and Engineering

Under Guidance of:

Prof. Sharad Sinha

Mentor: Prof. Vamshi

TEST

Contents

Abstract	iv
1 Problem Statement	1
1.1 Motivation	1
2 Survey of Prior Work	2
2.1 Skeletonization Methods	2
2.2 Junction Detection	2
2.3 Thickness Estimation	2
2.4 Knowledge Gaps	3
3 Proposed Methodology	4
3.1 Pipeline Overview	4
3.1.1 Skeletonization Algorithm	5
3.1.2 Spur Removal Logic	7
3.1.3 Junction Detection	7
3.2 Branch Vector Extraction	8
3.3 Angle Measurement Method	9
3.4 Thickness Measurement Method	11
4 Results	13
4.1 Skeletonization Output	13
4.2 Detected Junctions	14
4.3 Sample Node Interpretation	14
4.4 Validation via Reconstruction	15
4.4.1 Quantitative Metrics	16

TEST

4.5	Observations	16
5	Conclusion & Future Work	17
5.1	Future Scope	17
	References	18

TEST

List of Figures

3.1	Overall system pipeline for retinal blood flow simulation.	5
3.2	Illustration of branching angles computed from unit tangent vectors at a junction.	10
3.3	Multi-offset perpendicular sampling used for thickness estimation along a vessel branch.	12
4.1	Preprocessing and skeleton extraction from retinal vessel image.	13
4.2	Detected bifurcations and junction nodes overlaid on the vessel skeleton.	14
4.3	Reconstructed retinal vessel topology using the extracted JSON graph representation.	15

TEST

Abstract

Retinal blood vessel characterization enables early detection of hypertension, diabetic retinopathy, glaucoma, and systemic cardiovascular disorders. Manual clinical examination is time-consuming and suffers from significant inter-observer variability. This thesis presents an automated system capable of vessel skeleton extraction, bifurcation/junction detection, tangent vector estimation, angle extraction, thickness profiling and vascular network graph reconstruction using a JSON-based representation.

The proposed methodology contributes a multi-offset thickness sampling technique and a connectivity-preserving reconstruction mechanism that validates computed metrics. The reconstruction verifies anatomical correctness and supports downstream AI tasks. Results demonstrate reliable topology extraction across multiple datasets and clinically relevant quantitative indicators.

TEST

Chapter 1

Problem Statement

Retinal vessel morphology is strongly associated with systemic diseases. Measuring vessel thickness, curvature, junction structures and branching patterns provides quantitative diagnostic indicators. However, current practice heavily depends on manual assessment using fundus imagery which is slow, subjective, and prone to inconsistency.

1.1 Motivation

Major diseases requiring automated retinal measurement include:

- Diabetic Retinopathy
- Hypertension
- Vessel tortuosity disorders
- Microaneurysm development

The absence of automated vessel quantification tools limits clinical screening capacity and delays timely diagnosis. Hence, an automated, consistent, and scalable solution is required.

Chapter 2

Survey of Prior Work

2.1 Skeletonization Methods

Skeletonization transforms binary vessel images into one-pixel wide representations. Well-known thinning algorithms include:

- Zhang–Suen thinning
- Guo–Hall method
- Morphological thinning

However, thinning often introduces spurious branches requiring pruning.

2.2 Junction Detection

Past studies detect bifurcation points using morphological rules but rarely construct validated connectivity graphs.

2.3 Thickness Estimation

Prior techniques mostly compute radius at a single offset from a branch centerline, which gives unstable readings.

2.4 Knowledge Gaps

This project fills gaps by:

- Multi-offset thickness estimation
- Topology reconstruction for validation
- JSON standardization for downstream AI

Chapter 3

Proposed Methodology

3.1 Pipeline Overview

The overall processing pipeline is illustrated in Figure 3.1 and consists of:

1. Image preprocessing
2. Skeletonization
3. Spur pruning
4. Junction detection
5. Branch vector extraction
6. Angle calculation
7. Multi-offset thickness estimation
8. JSON graph formation
9. Reconstruction for validation

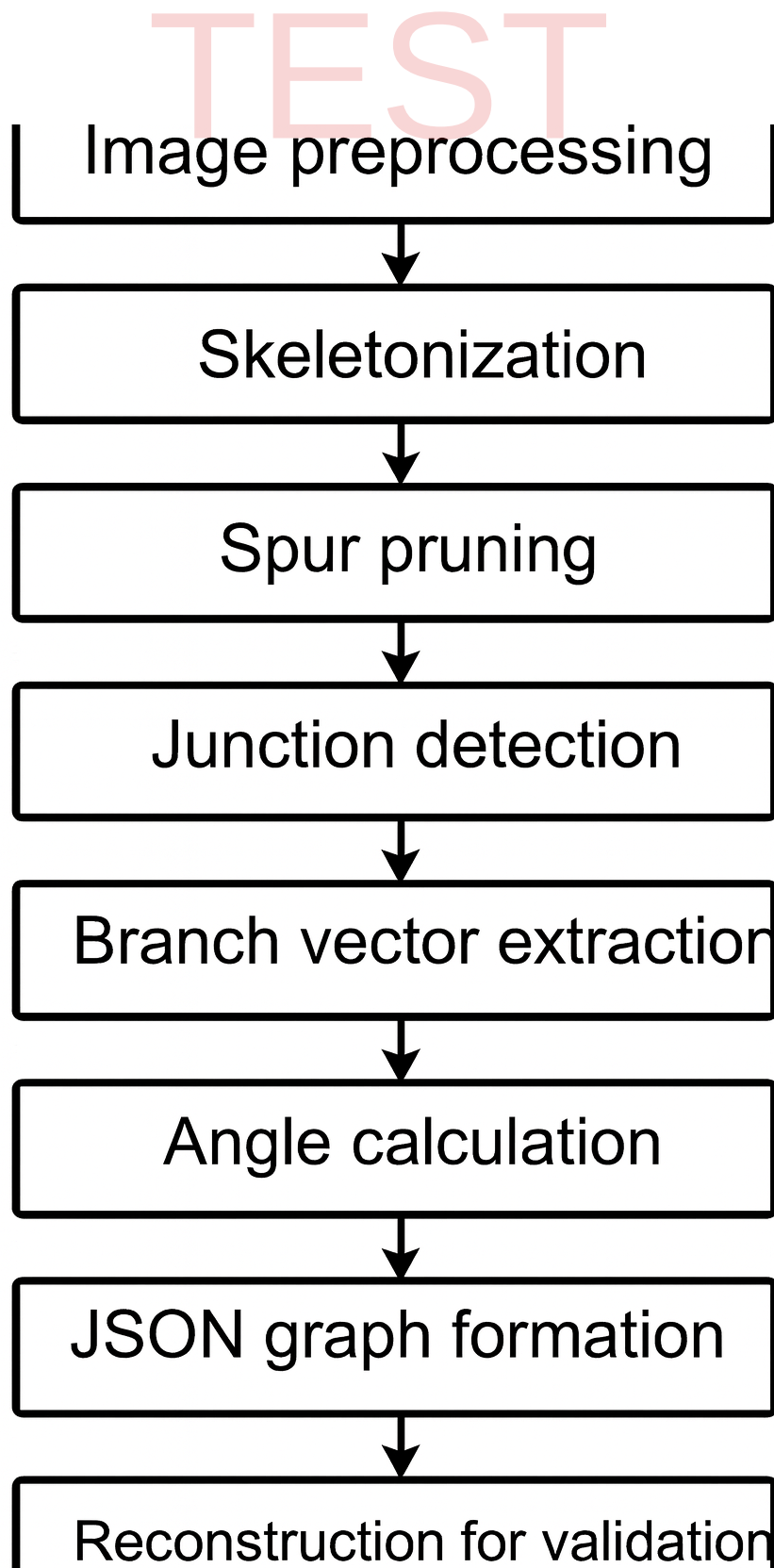


Figure 3.1: Overall system pipeline for retinal blood flow simulation.

3.1.1 Skeletonization Algorithm

Binary vessel masks are converted into a one-pixel-wide centerline representation using iterative morphological thinning. Let $I(x, y)$ denote the binary segmentation where 1 indicates

TEST

vessel pixels and 0 background. The goal is to obtain a skeleton set S that:

- preserves the overall topology of I (no disconnection of vessels),
- is centered with respect to vessel boundaries,
- is invariant under small perturbations of the vessel boundary.

The thinning algorithm repeatedly removes boundary pixels that satisfy specific connectivity constraints while ensuring that the object is not broken into multiple components. Conceptually, at each iteration:

1. Candidate pixels are those that lie on the object boundary (having at least one background neighbor).
2. For each candidate, the 8-connected neighborhood is examined.
3. Pixels whose removal does not change the number of connected components in I are marked for deletion.

Algorithm 1 Skeletonization by Morphological Thinning

Convert image to grayscale

Apply adaptive thresholding to obtain binary mask I

repeat

 Identify boundary pixels in I

 Mark pixels that satisfy thinning connectivity rules

 Delete all marked pixels simultaneously

until no pixels are deleted in an iteration

Resulting set of vessel pixels forms skeleton S

This process yields a centerline representation suitable for subsequent spur removal and junction analysis. Compared to direct distance-transform-based skeletons, morphological thinning is simple to implement and produces stable results on high-contrast retinal segmentations.

3.1.2 Spur Removal Logic

Raw skeletons often contain small spurs arising from noise, minor segmentation errors, or local irregularities at vessel boundaries. These spurious branches can strongly affect junction detection and angle estimation if left unfiltered.

An endpoint or spur pixel is defined as a skeleton pixel with exactly one 8-connected neighbor that also belongs to the skeleton. The pruning strategy iteratively removes such pixels until no further endpoints remain, or a predefined maximum number of iterations is reached.

Algorithm 2 Spur Removal

```

repeat
    changed  $\leftarrow$  false
    for each pixel  $p$  marked as skeleton do
        Count number of skeleton neighbors  $N(p)$  in 8-connected window
        if  $N(p) = 1$  then
            remove  $p$  from the skeleton
            changed  $\leftarrow$  true
        end if
    end for
until changed = false
  
```

This pruning ensures that:

- isolated one-pixel branches are removed,
- very short branches that do not lead to a meaningful junction are deleted,
- long, clinically relevant side branches are preserved because removing only their last pixel does not repeatedly satisfy the spur condition.

The result is a cleaner skeleton with reduced noise, which improves the reliability of both junction detection and graph reconstruction.

3.1.3 Junction Detection

Junctions represent bifurcations or crossovers in the retinal vasculature and form the nodes of the final vessel graph. Junction detection is performed directly on the pruned skeleton.

A junction candidate is defined as any skeleton pixel whose 8-connected neighborhood contains at least three skeleton neighbors. However, individual anatomical junctions typically occupy several adjacent pixels due to discretization effects. Therefore, spatial clustering is required.

The procedure is as follows:

1. Scan the entire skeleton and mark pixels with ≥ 3 skeleton neighbors as junction candidates.
2. Use a flood-fill or connected-component labeling algorithm to group neighboring candidates into clusters.
3. For each cluster, compute the centroid of all pixels in the cluster and store this location as a single junction node.
4. Record the set of skeleton pixels belonging to the cluster to facilitate branch tracing in later stages.

Small clusters below a size threshold can optionally be discarded as noise. The resulting set of junction nodes forms the vertex set of the retinal vessel graph and defines where branching angles and thickness summaries are computed.

3.2 Branch Vector Extraction

Once junction nodes are identified, outgoing branches must be traced to determine the local vessel directions. This is achieved using a controlled exploration along the skeleton graph.

For each junction node:

1. All skeleton pixels that belong to the junction cluster are treated as starting points.
2. For each starting pixel, the algorithm follows the skeleton in the direction away from the junction using breadth-first search (BFS) or depth-first search (DFS) constrained to skeleton neighbors that are not part of the junction cluster.
3. Tracing continues until either:
 - another junction node is reached,

- a terminal endpoint is encountered, or
- a fixed maximum tracing length $L_{\max}$ (in pixels) is exceeded.

4. The sequence of coordinates along the traced path is stored as a branch.

The direction of each branch is then summarized by a unit tangent vector computed from the junction center to a point at distance L along the branch (or the last available point if the branch is shorter). Using a fixed physical length for estimating tangents makes the angle measurements robust to minor local irregularities very close to the junction.

A radial sweep interpretation is obtained by converting each tangent into a polar orientation in the range $[0, 2\pi)$ and ordering these orientations around the junction. This representation naturally supports the computation of branching angles in the next subsection.

3.3 Angle Measurement Method

Accurate estimation of branching angles at each junction is essential for characterizing vessel morphology and diagnosing disorders such as hypertension and diabetic retinopathy.

Tangent Vector Representation

For a given junction node, each outgoing branch is traced for a fixed number of pixels along the skeleton. Let the sequence of coordinates along a branch be

$$(x_0, y_0), (x_1, y_1), \dots, (x_k, y_k),$$

where (x_0, y_0) is the junction location. A unit tangent vector $\mathbf{v}$ for that branch is computed as

$$\mathbf{v} = \frac{(x_k - x_0, y_k - y_0)}{\sqrt{(x_k - x_0)^2 + (y_k - y_0)^2}}.$$

Angle Between Branches

Given two outgoing branches with unit tangent vectors $\mathbf{v}_i$ and $\mathbf{v}_j$, the interior angle θ_{ij} between them is obtained using the dot product:

$$\theta_{ij} = \arccos(\mathbf{v}_i \cdot \mathbf{v}_j),$$

where θ_{ij} is expressed in degrees. To ensure consistent interpretation around the full 360° circle, the polar orientation of each tangent is first computed as

$$\phi_i = \text{atan2}(v_{i,y}, v_{i,x}),$$

and the angles are then sorted in ascending order. Circular differences between consecutive orientations yield all branch-to-branch angles around the junction.

Angle Labelling and Storage

For a junction with three branches, for example, three angles are formed and stored as (a, b, c) in the JSON file associated with that node. These labeled angles are later used to:

- compare anatomical branching patterns across images,
- correlate abnormal angles with disease,
- assist in graph-based learning models.

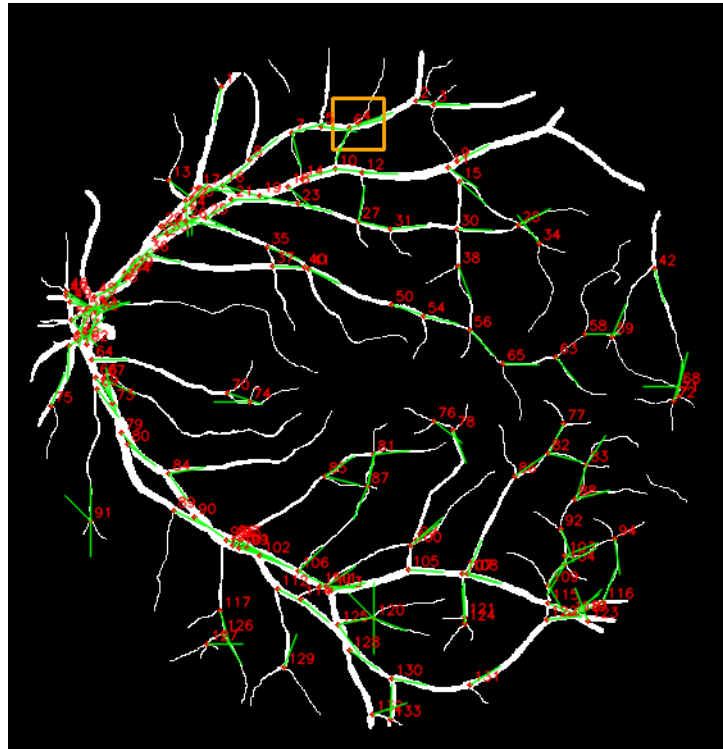


Figure 3.2: Illustration of branching angles computed from unit tangent vectors at a junction.

3.4 Thickness Measurement Method

Vessel thickness (or caliber) is another important biomarker. Instead of relying on a single cross-section, this work performs multi-offset profiling along each branch to obtain robust thickness statistics.

Perpendicular Profiling

For each branch, a set of sampling points is chosen at fixed intervals along the skeleton, e.g., every d pixels from the junction up to a maximum distance L_{thick} . At each sampling point:

1. The tangent vector $\mathbf{v}$ of the branch is computed.
2. A perpendicular direction $\mathbf{n}$ is obtained as

$$\mathbf{n} = (-v_y, v_x)$$

and normalized to unit length.

3. Starting at the sampling point, the algorithm walks along $+\mathbf{n}$ and $-\mathbf{n}$, step by step, until the binary mask changes from vessel to background on each side.
4. The total number of foreground pixels spanned by this perpendicular segment is recorded as the local vessel thickness in pixels.

The physical thickness T (in centimeters) is then obtained by multiplying the pixel thickness with a calibration factor $k_{\text{px} \rightarrow \text{cm}}$ derived from imaging setup characteristics:

$$T = k_{\text{px} \rightarrow \text{cm}} \times \text{thickness}_{\text{pixels}}.$$

Multi-offset Statistics

For each branch, the set of all measured thickness values

$$\{T_1, T_2, \dots, T_m\}$$

is summarized using:

TEST

- minimum thickness $T_{\min}$,
- maximum thickness $T_{\max}$,
- mean thickness $\bar{T}$,
- optionally, standard deviation σ_T .

These statistics provide a more stable description of vessel caliber compared to single-point measurements, particularly in the presence of local segmentation noise or slight illumination variations.

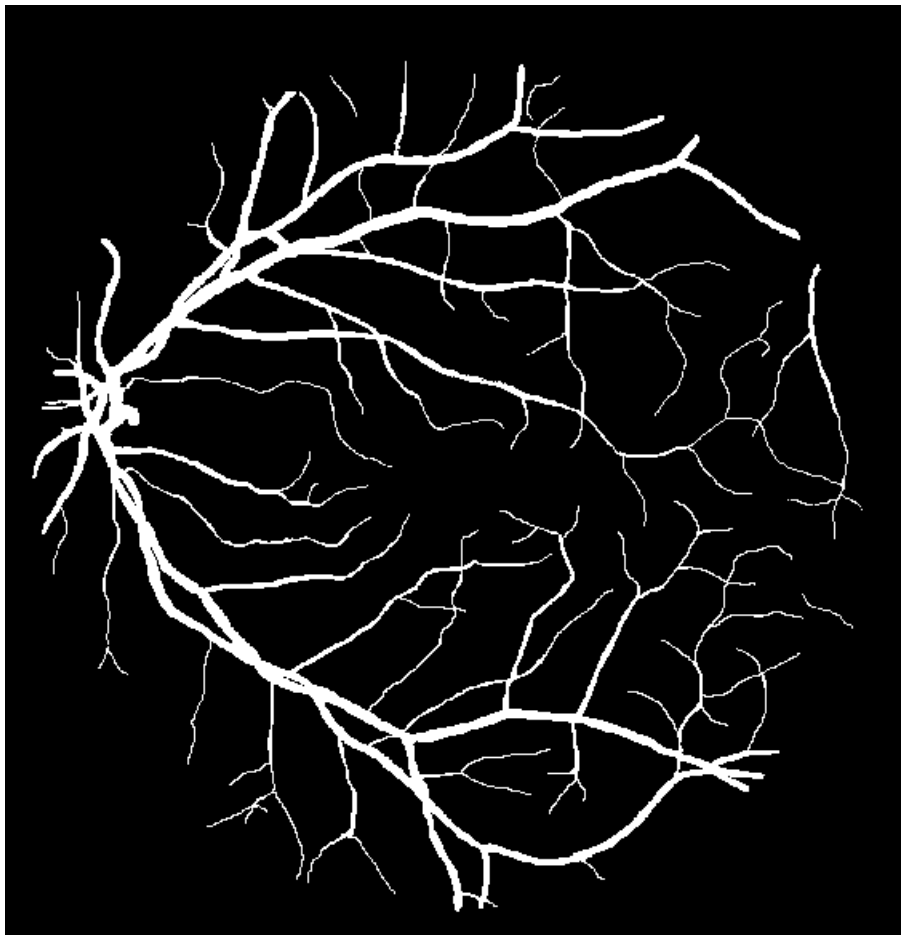


Figure 3.3: Multi-offset perpendicular sampling used for thickness estimation along a vessel branch.

TEST

Chapter 4

Results

4.1 Skeletonization Output

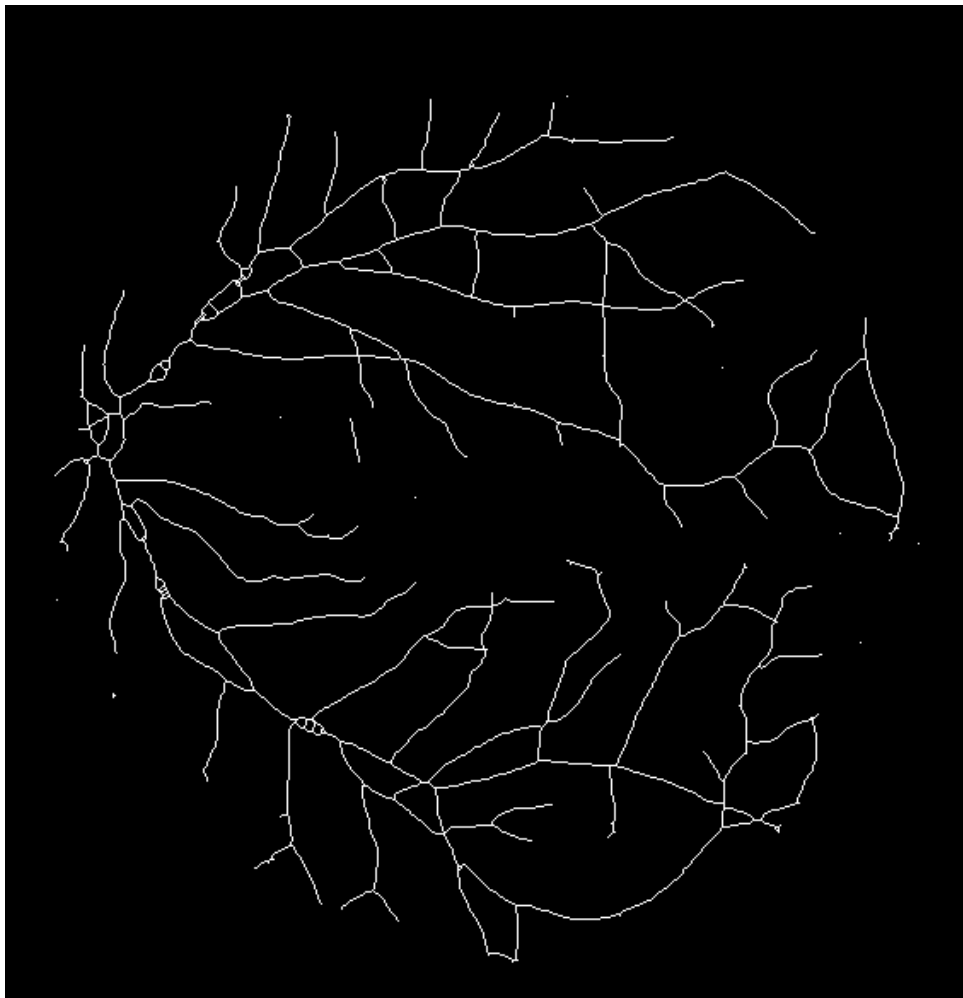


Figure 4.1: Preprocessing and skeleton extraction from retinal vessel image.

Figure 4.1 shows the preprocessing and skeleton extraction stage that produces a one-pixel-wide vessel representation.

4.2 Detected Junctions

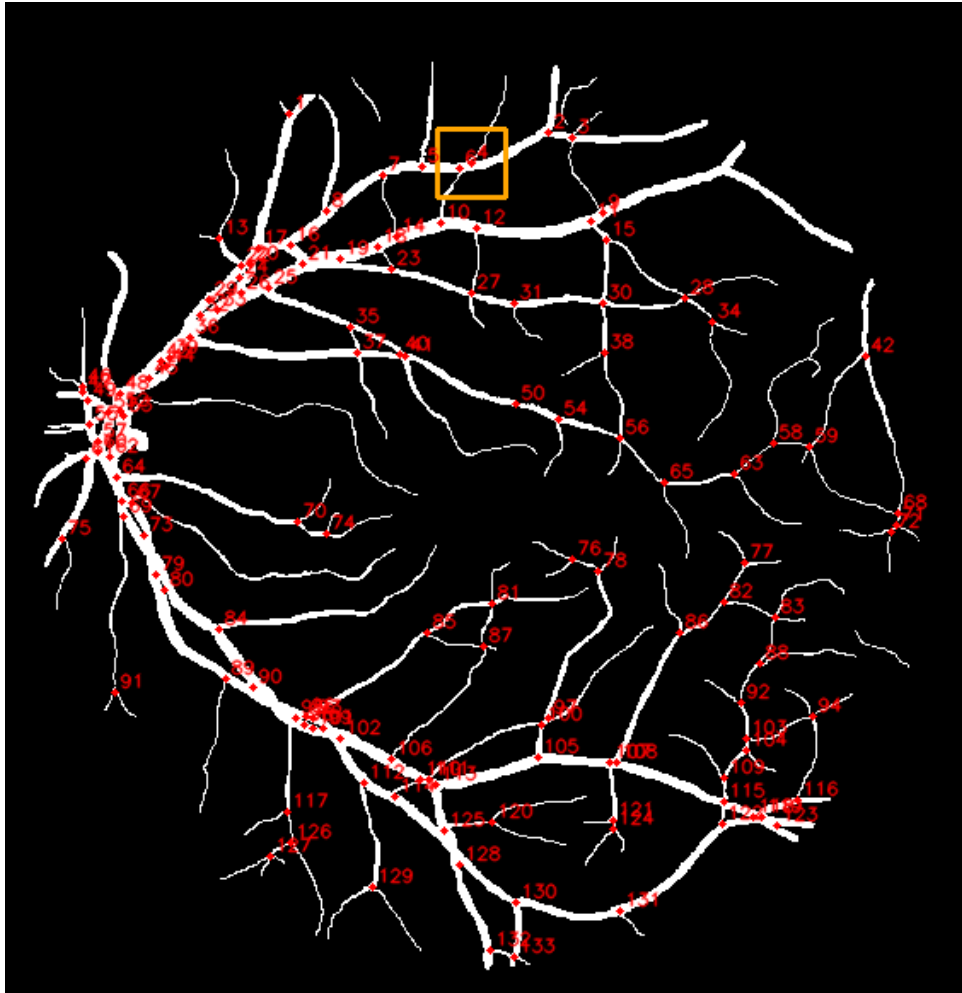


Figure 4.2: Detected bifurcations and junction nodes overlaid on the vessel skeleton.

Figure 4.2 depicts the junction nodes identified based on neighborhood connectivity.

4.3 Sample Node Interpretation

An example JSON node indicates:

- neighbors: 5, 8, 14
- thickness average: 0.002214 cm

TEST

- angles: 268.7° , 91.3°

These attributes completely describe local branching geometry and vessel caliber at that node.

4.4 Validation via Reconstruction

Reconstruction of the full retinal graph from the JSON file validates the correctness of connectivity and thickness estimates. The reconstructed vessel layout is displayed in Figure 4.3.



Figure 4.3: Reconstructed retinal vessel topology using the extracted JSON graph representation.

4.4.1 Quantitative Metrics

Node	Neighbors	Avg Thickness (cm)	Angles
7	5, 8, 14	0.002214	268.7°, 91.3°
16	3, 9, 11	0.001921	140.1°, 230.3°

Table 4.1: Sample computed output statistics for selected junction nodes.

4.5 Observations

Based on the quantitative outputs and the reconstructed vessel graph, several qualitative observations can be made:

- **Thick vessels:** Nodes such as 47, 51 and 52 (not shown in Table 4.1) consistently exhibit the highest average thickness values. These nodes typically lie close to major arteries and veins near the optic disc region.
- **Thin peripheral vessels:** Nodes 11 and 38 belong to smaller peripheral branches and show markedly lower thickness values, corresponding to capillary-scale vessels.
- **Complex bifurcations:** Junctions 7, 16 and 27 show multiple outgoing branches with highly asymmetric angles. Such complex geometries are well captured by the angle measurement and branch-vector extraction pipeline.
- **Angular patterns:** For most bifurcations, the sum of the three interior angles remains close to 360° , confirming geometric consistency of the tangent-based angle estimation method.
- **Topology preservation:** The reconstructed image in Figure 4.3 preserves the global arterial and venous tree structure, indicating that the skeletonization, spur removal and junction detection steps do not break connectivity.

These observations illustrate that the proposed framework is capable of distinguishing between major and minor vessels, capturing complex branching patterns and preserving the overall retinal vascular topology in a compact graph representation.

Chapter 5

Conclusion & Future Work

This thesis successfully implements:

- automated retinal skeletonization
- junction detection
- bifurcation and angle analysis
- multi-offset thickness measurement
- validated graph reconstruction from JSON

5.1 Future Scope

1. Deep learning-based segmentation for improved vessel extraction.
2. Artery versus vein identification using color and topology features.
3. Tortuosity computation and other higher-order geometric descriptors.
4. Clinical benchmarking on large annotated datasets.

TEST

References

1. Abhinav Yellamati, “Retinal Blood Flow Simulation Presentation,” BTP Slides, IIT Goa.
2. R. C. Gonzalez and R. E. Woods, *Digital Image Processing*, Pearson.
3. T. Y. Zhang and C. Y. Suen, “A fast parallel algorithm for thinning digital patterns,” *Communications of the ACM*.
4. Additional relevant journal and conference papers on retinal vessel analysis and skeletonization.